

Gaurav Khanal

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PERSONAL SUMMARY

M2 Data Science & AI student focused on scientific machine learning, physics-informed modeling, and interpretable topology- and geometry-aware AI for physical systems. Research experience includes hyperelastic material identification, predictive constitutive modeling, inverse problems, topological data analysis, signal processing and uncertainty quantification.

EDUCATION

MSc in Data Science and Artificial Intelligence Sep. 2024—Aug. 2026
Université Côte d'Azur, Sophia Antipolis, France
EFELIA Bourse d'Excellence 2025 (*Scholarship of Excellence*)

BA in Economics and Accounting Aug. 2009—May 2013
Westminster College, Fulton, Missouri, USA

PROFESSIONAL EXPERIENCE

Research Intern, Predictive Constitutive Modeling of Atrial Appendage Tissue Apr. 2026—Present
Mines Saint-Étienne | Centre Ingénierie et Santé | SAINBIOSE Lab Inserm U1059

- Developing mechanics-informed ML models to infer specimen-specific force response and constitutive behavior of atrial appendage (AA) tissue from projected 2D thickness maps and 3D surface mesh scans.
- Engineering 3D morphology and anisotropy descriptors from tissue meshes, including loading-relative geometry, curvature morphology, Laplace–Beltrami spectral signatures, principal-curvature ridge/saddle/cap-cup features, and fabric-tensor anisotropy.
- Extending the framework with point-cloud and mesh representation learning, including PointNet++ and mesh GNN baselines, to predict mechanical properties and constitutive parameters from 3D atrial tissue morphology.
- Investigating topology-aware descriptors for 3D tissue scans using topological data analysis, including persistent-homology-based summaries of multiscale shape, fold, ridge, and cycle structure, to assess whether topological features improve prediction of mechanical properties and constitutive parameters beyond geometric and graph-based baselines.
- Developing hybrid mechanics-informed models that map learned 2D/3D tissue representations to constitutive-law parameters, with force-response curves generated through differentiable Veronda–Westmann (isotropic) and/or Holzapfel–Gasser–Ogden (anisotropic) constitutive layers.

RESEARCH EXPERIENCE

Physics-Informed Image Registration for Hyperelastic Material Identification Jan.—Apr. 2026

- Developed a unified PINN framework for soft-tissue characterization that couples differentiable image registration with hyperelastic material-parameter identification, replacing the classical sequential digital image correlation (DIC) → displacement extraction → constitutive fitting pipeline.
- Integrated Ogden-model equilibrium constraints, explicit gravity loading, boundary-condition penalties, and multi-step reaction-force losses to regularize deformation fields and improve identifiability of material parameters.
- Generated synthetic tissue image sequences replicating the Gao–Desai porcine liver experiment, including gravity-induced pre-compression, tensile loading, natural-texture synthesis, and realistic imaging noise.
- Recovered Ogden parameters on synthetic validation with relative errors of 0.02% for μ and 0.006% for α , while implicitly resolving the zero-strain reference-state ambiguity that classical DIC pipelines estimate heuristically.

Deep Learning for Full Waveform Inversion: Seismic Waveforms to Velocity Fields May—Aug. 2025
Supervisor: Dr. Micheil Riveill, INRIA/I3S Lab, Université Côte d'Azur

- Developed a data-driven FWI framework to learn the inverse map from multi-source seismic shot gathers to subsurface velocity fields, targeting key limitations of conventional FWI: high computational cost, ill-posedness, and cycle skipping.
- Designed a hybrid Swin V2 Transformer + enhanced U-Net architecture combining long-range wavefield dependency modeling with high-fidelity spatial reconstruction of geological layers, faults, and heterogeneous structures.
- Built a scalable HPC OpenFWI training pipeline for 470,000 Vel/Fault/Style samples using memory-mapped data loading, GPU augmentations, PyTorch DDP/AMP, EMA checkpointing, and H100 GPUs.

- Formulated a custom loss with inverse frequency weighting, and achieved best validation MAE of 63.27 m/s, with SSIM 0.8725-0.9188 across OpenFWI families; errors were concentrated mainly near sharp interfaces and fault boundaries.

Graph Signal Processing for Characterization of Multipolar Electrograms in AF

Oct. 2025—Mar. 2026

Supervisor: Dr. Vicente Zarzoso, I3S Lab, Université Côte d'Azur

- Developed a geometry-aware graph signal processing framework for multipolar intracardiac electrograms (EGMs) in Atrial Fibrillation (AF), modeling PentaRay catheter electrodes as graph nodes and EGM recordings as graph signals over catheter geometry.
- Applied joint Short-Term Fourier Transform and Graph Fourier Transform analysis and graph total variation to detect rotor-proximal spatial discordance, showing that AF-band graph-time variation captured intermittent wavefront-collision patterns missed by per-channel temporal features.
- Proposed edge-flow extensions of catheter graph signals, enabling propagation dynamics to be interpreted through discrete differential operators and motivating Helmholtz-Hodge-style separation of planar/focal activation, rotor-like circulation and obstacle-related harmonic flow, providing a mechanistic route toward rotor-proximity indicators for AF ablation guidance.
- Achieved 87.5% accuracy and $AUC = 1.00$ for close-vs-far rotor classification on a simulated OpenCARP proof-of-concept dataset, outperforming conventional time-domain and phase-mapping baselines.

PERSIST: Topology-Guided Multiscale Domain Identification in Spatial Transcriptomics

Sep.—Dec. 2025

- Developed PERSIST, an unsupervised topology-guided framework for spatial transcriptomics domain identification, addressing resolution sensitivity by inferring intrinsic tissue scales rather than relying on fixed clustering resolutions, smoothing radii, or neighborhood parameters.
- Constructed Alpha-complex filtrations over tissue coordinates and used H_1 persistent homology birth-time distributions to identify topology-informed spatial scales associated with compartmentalization and boundary structure.
- Designed a Discrete Exterior Calculus-based multiscale stability field using 0-form smoothing, discrete gradients, topology-weighted local variation, bootstrap robust-core estimation, and Morse-like basin decomposition to separate stable domains, transition regions, and high-gradient boundaries.
- Applied PERSIST to breast cancer, human lymph node, and mouse brain spatial transcriptomics datasets, recovering biologically interpretable tissue organization supported by marker enrichment and H&E histology overlays.

SKILLS

Computer Languages: Python, R, MATLAB

Machine / Deep Learning Frameworks: PyTorch, JAX, Equinox, Flax, TensorFlow, scikit-learn

Scientific Computing & Geometry: NumPy, pandas, SciPy, CuPy, Gudhi, Trimesh, PyVista, ParaView

HPC, MLOps & Development: Grid5000, OAR, CUDA, PyTorch DDP, Git, Docker, MLflow, TensorBoard

Languages: English (C2—*fluent*), French (B1—*intermediate*), Spanish (A2—*elementary*)

CONTRIBUTIONS & COMPETITIONS

TopoBench TDL Challenge 2026

- Built and evaluated $E(n)$ -Equivariant Topological Neural Network (ETNN) variants for graph learning, extending graph data into combinatorial complexes and implementing coordinate-free and Laplacian Positional Encoding (LapPE) structural-coordinate backbones with tests and full GraphUniverse benchmark evaluation.

OTHER PROFESSIONAL EXPERIENCE

Senior Consultant, Ernst & Young (Melbourne, Australia)

Jul. 2021 – Apr. 2024

Senior Consultant, Protiviti (USA & Australia)

Jul. 2017 – Jun. 2021

Experienced Consultant, PricewaterhouseCoopers (Kansas City, Missouri, USA)

Sep. 2013 – Jul. 2017

REFERENCES

Dr. Baptiste Pierrat (Current Supervisor)

pierrat@emse.fr

Research Engineer at SAINBIOSE Lab and Professor at Mines Saint-Étienne

Dr. Vicente Zarzoso (Former Supervisor and Professor)

vicente.zarzoso@univ-cotedazur.fr

Professor and Researcher, I3S Lab, Université Côte d'Azur

Dr. Michel Riveill (Former Supervisor and Professor)

michel.riveill@inria.fr

Professor and Program Director, MSc Data Science & AI, Université Côte d'Azur, & Researcher at INRIA